SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED
CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)
Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given

subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1

Given

- Network Address: **192.168.20.0/24**
- Total addresses available = **256 (0–255)**
- Three labs are assigned different subnet masks:
    - Lab A → **/26**
    - Lab B → **/27**
    - Lab C → **/28**

Step 1: Lab A (/26)

Subnet Mask:

255.255.255.192

Block Size:

256 − 192 = 64

The subnet starts from **192.168.20.0**.

- **Network Address:** 192.168.20.0
- **First Host:** 192.168.20.1
- **Last Host:** 192.168.20.62
- **Broadcast Address:** 192.168.20.63

Number of Addresses:

2^(32−26)

= 2^6

= 64 addresses

Usable Hosts:

64 − 2

= 62 hosts

Step 2: Lab B (/27)

Next available subnet begins after Lab A.

Network Address:

192.168.20.64

Subnet Mask:

255.255.255.224

Block Size:

256 − 224 = 32

Therefore,

- **Network Address:** 192.168.20.64
- **First Host:** 192.168.20.65
- **Last Host:** 192.168.20.94
- **Broadcast Address:** 192.168.20.95

Number of Addresses:

2^(32−27)

= 2^5

= 32

Usable Hosts:

32 − 2

= 30 hosts

Step 3: Lab C (/28)

Next available subnet:

192.168.20.96

Subnet Mask:

255.255.255.240

Block Size:

256 − 240 = 16

Therefore,

- **Network Address:** 192.168.20.96
- **First Host:** 192.168.20.97
- **Last Host:** 192.168.20.110
- **Broadcast Address:** 192.168.20.111

Number of Addresses:

2^(32−28)

= 2^4

=16

Usable Hosts:

16−2

=14 hosts

Final Table

Lab	Subnet	Network Address	First Host	Last Host	Broadcast	Usable Hosts
A	/26	192.168.20.0	192.168.20.1	192.168.20.62	192.168.20.63	62
B	/27	192.168.20.64	192.168.20.65	192.168.20.94	192.168.20.95	30
C	/28	192.168.20.96	192.168.20.97	192.168.20.110	192.168.20.111	14

Conclusion

The /24 network is successfully divided into three fixed-size subnets. Lab A can support **62 hosts**, Lab B **30 hosts**, and Lab C **14 hosts**.

Question 2

Given

System 1

IP Address = 192.168.1.130

Subnet Mask = 255.255.255.192 (/26)

System 2

IP Address = 192.168.1.190

Subnet Mask = 255.255.255.192 (/26)

Step 1: Determine Block Size

Subnet Mask:

255.255.255.192

Block Size

256−192=64

Possible subnets are

Subnet	Range
192.168.1.0	0–63

Subnet	Range
192.168.1.64	64–127
192.168.1.128	128–191
192.168.1.192	192–255

System 1

IP Address

192.168.1.130

130 falls between

128–191

Hence,

- Network Address = **192.168.1.128**
- First Host = **192.168.1.129**
- Last Host = **192.168.1.190**
- Broadcast = **192.168.1.191**

Verification

130 lies between

129 and 190

Therefore,

The IP address belongs to this subnet.

Usable Hosts

$2^{(32-26)}=64$

$64-2=62$

System 2

IP Address

192.168.1.190

190 also falls within

128–191

Therefore,

- Network Address = **192.168.1.128**
- First Host = **192.168.1.129**
- Last Host = **192.168.1.190**
- Broadcast = **192.168.1.191**

Verification

190 is the last usable host.

Therefore,

The IP belongs to the subnet.

Usable Hosts

62

Final Table

System	IP	Network	Host Range	Broadcast	Valid	Usable Hosts
1	192.168.1.130	192.168.1.128	129–190	191	Yes	62
2	192.168.1.190	192.168.1.128	129–190	191	Yes	62

Conclusion

Both systems belong to the same subnet **192.168.1.128/26**, and each subnet provides **62 usable host addresses**.

Question 3

Given

Network

172.16.0.0/16

Departments

- HR → /25
- Sales → /26
- IT → /27
- Admin → /28

HR Department (/25)

Subnet Mask

255.255.255.128

Block Size

128

Network

172.16.0.0

Host Range

172.16.0.1

to

172.16.0.126

Broadcast

172.16.0.127

Usable Hosts

2^7=128

128−2=126

Sales Department (/26)

Next subnet

172.16.0.128

Host Range

172.16.0.129

to

172.16.0.190

Broadcast

172.16.0.191

Usable Hosts

2^6=64

64−2=62

IT Department (/27)

Next subnet

172.16.0.192

Host Range

172.16.0.193

to

172.16.0.222

Broadcast

172.16.0.223

Usable Hosts

30

Admin Department (/28)

Next subnet

172.16.0.224

Host Range

172.16.0.225

to

172.16.0.238

Broadcast

172.16.0.239

Usable Hosts

14

Remaining Unused Address Range

Allocated addresses end at

172.16.0.239

Unused addresses begin from

172.16.0.240

Remaining range

172.16.0.240

to

172.16.255.255

Final Table

Department	Network	Host Range	Broadcast	Usable Hosts
HR	172.16.0.0/25	172.16.0.1–172.16.0.126	172.16.0.127	126
Sales	172.16.0.128/26	172.16.0.129–172.16.0.190	172.16.0.191	62
IT	172.16.0.192/27	172.16.0.193–172.16.0.222	172.16.0.223	30
Admin	172.16.0.224/28	172.16.0.225–172.16.0.238	172.16.0.239	14

Conclusion

The departments are allocated consecutive subnets within the **172.16.0.0/16** network. After assigning the required subnets, the remaining address space from **172.16.0.240 to 172.16.255.255** remains available for future use.

Question 4

Given

Network

172.16.10.0/24

Subnet Mask

/27

Step 1: Total Number of Subnets

Borrowed bits

27−24=3

Number of subnets

2^3=8

Step 2: Number of Hosts

Host bits

32−27=5

Total addresses

2^5=32

Usable hosts

32−2=30

Step 3: List of Subnets

Subnet	Range
172.16.10.0	0–31
172.16.10.32	32–63
172.16.10.64	64–95
172.16.10.96	96–127
172.16.10.128	128–159
172.16.10.160	160–191
172.16.10.192	192–223
172.16.10.224	224–255

Step 4: Determine the Subnet of 172.16.10.78

The address

172.16.10.78

falls in

64–95

Hence,

Network Address

172.16.10.64

Host Range

172.16.10.65

to

172.16.10.94

Broadcast

172.16.10.95

Step 5: Check 172.16.10.95

The address

172.16.10.95

is the **broadcast address** of this subnet.

Therefore,

- It belongs to the same subnet.
- It **cannot** be assigned to any host.

Conclusion

- Total Subnets = **8**
- Usable Hosts per Subnet = **30**
- IP **172.16.10.78** belongs to subnet **172.16.10.64/27**.
- Network Address = **172.16.10.64**
- Broadcast Address = **172.16.10.95**
- IP **172.16.10.95** is in the same subnet but is reserved as the broadcast address.

Question 5

Given IPv6 Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1: Compress the IPv6 Address

IPv6 compression rules:

1. Remove leading zeros in each block.
2. Replace one continuous sequence of all-zero blocks with "::".

Compressed address

2001:db8::ff00:42:8329

Step 2: Expand the Address

Expand every block to four hexadecimal digits.

Expanded address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine the Network Prefix (/64)

A **/64 prefix** means the first **64 bits** (first four hexadecimal blocks) represent the network.

Network Prefix

2001:db8:0:0::/64

or equivalently

2001:db8::/64

Step 4: Calculate the Number of Host Addresses

Host bits

128−64=64

Possible host addresses

2^64

18,446,744,073,709,551,616

Final Table

Item	Value
Original Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed Address	2001:db8::ff00:42:8329
Expanded Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix Length	/64
Network Prefix	2001:db8::/64
Host Bits	64
Possible Host Addresses	2^{64} = 18,446,744,073,709,551,616